DUAL-ARM ROBOTIC CUTTING

A beginner guide to the work so far

What was built. What was tested. What failed. What is still a plan.

Prepared for the project owner | 21 October 2026

CURRENT POSITION

Established in its stated scope	Not established
Historical scripted lifts and local-resource feasibility	A trained camera-based knife-pickup policy
Offline deformable tomato and contact checks	A complete physical slice with independent release
Latest 103-test suite and triangle-contact repair	Real-robot cutting or calibrated tomato material

This report explains the entire documented progression in plain language. Historical demonstrations, conditional calculations, numerical references and future research steps are kept separate. It is a record of project evidence, not a certification of hardware readiness.

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Read chapters 1-5 for the overall picture, 6-13 for the robot and grasp work, 14-26 for tomato mechanics, and 27-34 for criteria, next steps and your own checks.

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1 What this project is trying to achieve

The long-term goal is a dual-arm robot that can find a kitchen knife, pick it up firmly, locate a tomato, stabilize it, and produce slices. Eventually the same ideas should work on the physical robot. Development is restricted to your laptop; cloud training is outside the agreed scope.

We first explored the hand and knife. When memory limited further simulator testing, the work shifted to the tomato itself: how to deform it, represent damage, prevent impossible overlap, and tell whether a slice has truly separated.

Area	Current position
Scripted knife demonstrations	Limited arranged-scene lifts were recorded. They are historical baselines.
Knife pickup from a flat board centre	A firm wrapped pickup has not passed the frozen criteria.
Vision-based RL pickup	Architecture and interfaces planned; no trained recognition or pickup policy.
Tomato mechanics	A restricted offline spring/volume prototype and many numerical checks exist.
Complete tomato cut	Not validated. Deeper contact solves remain a gate.
Physical robot deployment	Not performed; interfaces, sensors and limits remain unverified.

The most important distinction: code running, a test passing, and the robot accomplishing the final research task are three different things.

This report is a consolidated explanation of the saved workspace records through 21 October 2026. Earlier evidence is drawn from its reports and saved outputs; preparing this PDF does not rerun those historical experiments. No simulator, training or robot is launched to make the report.

Evidence: [S01], [S02], [S03], [S37]. See the source index.

2 How to judge the evidence

Four kinds of work

- A plan says what should happen later. It does not establish that a feature exists.
- An offline check runs calculations on saved geometry or equations without starting Isaac Sim.
- A simulator experiment advances a physics scene. In the records, "physical grasp test" means simulated contact physics, not a real robot trial.
- A hardware experiment would move the real robot or use its live sensors. None is established here.

Why a green test is not the whole answer

A unit test checks one rule, such as "damage must not decrease." A coupon is a small, controlled mechanical example. An integration check tests several parts together. A full task demonstration needs all relevant parts to work under the declared conditions.

For example, deleting every connecting bond can produce two graph components. That proves the connectivity checker works. It does not prove that a blade physically broke those bonds, that the pieces can collide correctly, or that a hand can carry the knife.

Label used here	Meaning
Demonstrated in scope	The stated experiment passed its declared checks.
Conditional / provisional	The result depends on unverified geometry, loads or material assumptions.
Failed / stopped	A required check failed, the worker stopped, or the resource guard refused execution.
Planned / not validated	There is no accepted result for that claim.

A normal program exit is not a success certificate. Conversely, a deliberate guard refusal or time-limit stop is not proof that the laptop crashed.

Evidence: [S03], [S20], [S36], [S37]. See the source index.

3 The project history at a glance

Stage	What changed	Main lesson
Scene and controls	Robot-only control, kitchen restoration and knife approach.	Reduce the scene and protect memory.
Scripted grasp	Fingertip pinch, then wrapped-grip and cutting-orientation baselines.	A small lift does not automatically mean a good grasp.
Vision/RL preparation	Interface audit, camera resource probe and native environment checks.	A framework working is not a learned pickup skill.
Central grasp development	Flat reset, tilting, palm seating and finger-contact transfer.	Supported handle rise is not unsupported retention.
Offline grasp review	Geometry, colliders, controller replay and load-support proposals.	Near contact and plausible pose are not force balance.
Tomato construction	Nodes, bonds, patch damage, closed faces and mass/stiffness.	Energy and geometry need their own checks.
Mechanical refinement	Volume protection, mesh repairs and sensitivity studies.	A converged numerical solve can still be physically invalid.
Contact/fracture	Contact coupons, coupled static solves and partial damage.	Breaking material and preventing penetration are separate duties.
Latest stage	Unloading, incomplete full-cut attempt and triangle contact repair.	The contact defect is fixed in scope; full cutting remains unproven.

Much of the saved design and test evidence is dated 19-20 October 2026. The report follows dependency order rather than inventing a precise timestamp for every individual command. Older scene demonstrations are retained as historical evidence, not silently promoted to current training results.

Evidence: [S01], [S03], [S17], [S37]. See the source index.

4 The robot, laptop and sensors

The right arm has seven independent joints. The right hand has six independent commands and five passive follower joints. Four fingers each have one main bending command; the thumb has yaw and pitch. Some smaller joints move through authored coupling rules rather than receiving separate commands.

A human can independently shape many finger joints. This robot hand cannot copy every human posture or closing sequence exactly. A useful target is a functional wrap that supports and retains the handle.

Recorded item	What it means
Finger coupling	Distal finger angle = 0.89 times the main finger angle.
Thumb coupling	Thumb follower angle = 1.86 times thumb pitch.
CPU / GPU	Audit recorded Intel Core i7-14650HX and RTX 5060 Laptop GPU.
Memory	About 16 GB host RAM and 8 GB VRAM were recorded. Available memory changes with other applications.
Sensor status	Real tactile configuration is unknown. Wrist torque is only a provisional slicing assumption.

The installed simulation metadata recorded Isaac Sim 6.1.0.0, an editable Isaac Lab checkout, Python 3.12.14 and a CUDA-capable PyTorch build. Isaac Lab distribution metadata and the checkout top-level version differ; the audit records both instead of treating a single label as a compatibility guarantee.

Before real robot use, the exact controller API, encoder order, signs, units, firmware, torque/speed limits and sensor timestamps must be verified. The hand SDK command order also needs named mapping: copying array positions could swap the thumb channels.

Evidence: [S02], [S04], [S06], [S13]. See the source index.

5 Why memory protection matters

RAM is the laptop memory shared by applications. VRAM is memory on the GPU. Loading a kitchen, robot, collision shapes, camera buffers and learning software can use both. Having free disk space does not make up for exhausted RAM or VRAM.

Workload	Recorded guard or limit
Older guarded scripted demos	Approximately 5 GB available RAM required at startup; host hard stop near 2 GB.
Later RL/native benchmark	10 GB available RAM and 4 GiB free VRAM at startup.
Later simulator reserves	Checkpoint reserve 3,000,000 Linux kB; hard stop 2,000,000 kB. GPU reserves 1.5 / 1 GiB.
Latest offline mechanical workers	512 MiB address space, 45 s CPU, 55 s wall alarm; one BLAS thread and a shared lock.

The historical launchers have different startup budgets. The later 10 GB requirement was based on measured simulator use and reserve, not a claim that every earlier launcher uses the same threshold. These protections were preserved rather than reduced to make a test start.

A shared lock prevents two robot-controller or mechanical workers from running at once. A checkpoint stores recoverable software state. A watchdog monitors resources and can stop a process group. They reduce risk; they cannot guarantee that every application or driver will never crash.

Memory limit versus time limit

The latest deeper offline contact probe hit a 45-second CPU cap. That is a solver-computation problem under a bounded worker budget, not evidence that it needed more RAM. More RAM can help simulator headroom, but it does not fix bad geometry, wrong contact equations or slow convergence.

Evidence: [S01], [S05], [S06], [S37]. See the source index.

6 Restoring the scene and moving the arm

Early work brought the robot, kitchen and the kitchen asset's existing knife into a usable scene. The knife was not introduced as a separate replacement asset. Scene arrangements were applied in disposable layers or memory so the source robot and kitchen files could be preserved.

A smaller robot-only setup was used to test joint control and physics with less memory. This separated basic articulation issues from the cost of a detailed kitchen. The right arm could then be commanded through joint drives while the left arm was positioned to clear the approach.

How reaching was programmed

The controller read the settled handle pose directly from the simulator. Inverse kinematics, or IK, calculated joint angles that put the palm near a desired position. A sequence of targets moved the hand toward the knife. This is programmed motion using privileged simulator information.

The first finite approach validation reported about 0.24 mm palm target error and about 120 mm clearance above the handle, with no detected right-arm contact. Those numbers show an approach target could be reached in that setup. They do not show grasping or vision recognition.

Scene housekeeping requested later

The nearer-grip experimental arrangement aligned the board with the countertop, deactivated the spare knife in the session and moved the board/knife arrangement to an accessible edge. The record says those scene edits were checked. It does not say the nearer grip then passed the full wrap-and-hold test.

No real robot was moved in this work. Looking through a rendered camera is also different from the robot detecting the knife from camera pixels.

Evidence: [S01], [S02]. See the source index.

7 From fingertip pinch to wrapped grip



Historical simulator capture: arranged-scene scripted cutting grip, visible trial 19. This is not a learned pickup or a tomato-cutting result.

The initial successful pinch used fingertips and lifted the free kitchen knife about 29.61 mm. It held for three seconds with about 0.385 mm relative slip. This was a useful baseline, but it was explicitly not the human-like enveloping grip you requested.

A later scripted baseline added palm/opposed-thumb support and at least three curled fingers. At least two qualifying contacts had to be on proximal/inner regions or sufficiently far from the fingertips. A lift alone could not pass the wrapped-grip check.

Recorded trial	Lift	Relative slip	Scope
18, headless	57.53 mm	1.75 mm	Three-second arranged-scene wrapped hold
19, visible	59.22 mm	0.07 mm	Same stated validation scope

The knife remained a free rigid body after physics began: no attachment or in-episode commanded knife pose was used. Friction and compliance were provisional, and this older demo had self-collision disabled. Its success must not be transferred to the later flat-centre, self-colliding scene.

Evidence: [S01], [S07]. See the source index.

8 Orientation, closing order and varied positions

You asked for the handle toward the robot, blade pointing outward and a firmer grip closer to the blade. In the recorded cutting-grip baseline, the hand lifted and then reoriented the blade outward with its edge toward the board.

The closure sequence began with fingers supporting the handle while the thumb remained open. After two fingers exceeded 0.1 N support for 0.15 seconds, firm thumb closure began. The index was delayed by one second relative to the other fingers. Incidental stationary-thumb touch was logged separately; this was not proof that every finger fully clamped before any thumb contact.

Historical placement offset	Outcome
Original location	Wrapped hold passed.
+5 cm world x	Inner wrap not retained throughout the hold.
-5 cm world x	Controlled stop: descent waypoint unreachable.
+5 cm world y	Wrapped hold passed.
-5 cm world y	Wrapped hold passed.

Three of five historical placements passed in the earlier grip orientation, with the board and knife translated together and the handle overhang retained. This was one trial per placement. It was not arbitrary knife recognition, a repeatability estimate, a rotated-handle study or a flat central pickup result.

Why the arm appeared slow

The motion was deliberately staged and bounded by controller targets, pressure acquisition, settling and hold checks. Rendering and solver computation also affect wall-clock speed. A slow visible demonstration alone cannot identify one cause. Simulator seconds, controller update rate and real elapsed time must be measured separately before changing speed.

Evidence: [S01], [S05]. See the source index.

9 What learning and vision would add

You changed the objective from a programmed demonstration to a learned, camera-based knife pickup skill. That is a different system: the robot should find the knife from images and choose useful actions across varied valid placements, rather than receive the exact handle location as the answer.

Planned stage	Beginner explanation
RGB-D camera	Colour plus depth helps estimate where the board, blade and handle are.
Perception model	Turns camera information into features or an estimated handle pose and confidence.
RL policy	Maps observations to bounded arm/hand actions, improving through evaluated experience.
Low-level controller	Turns commands into motor targets while enforcing limits.
Independent evaluator	Checks actual grasp quality, lift, slip and failures.

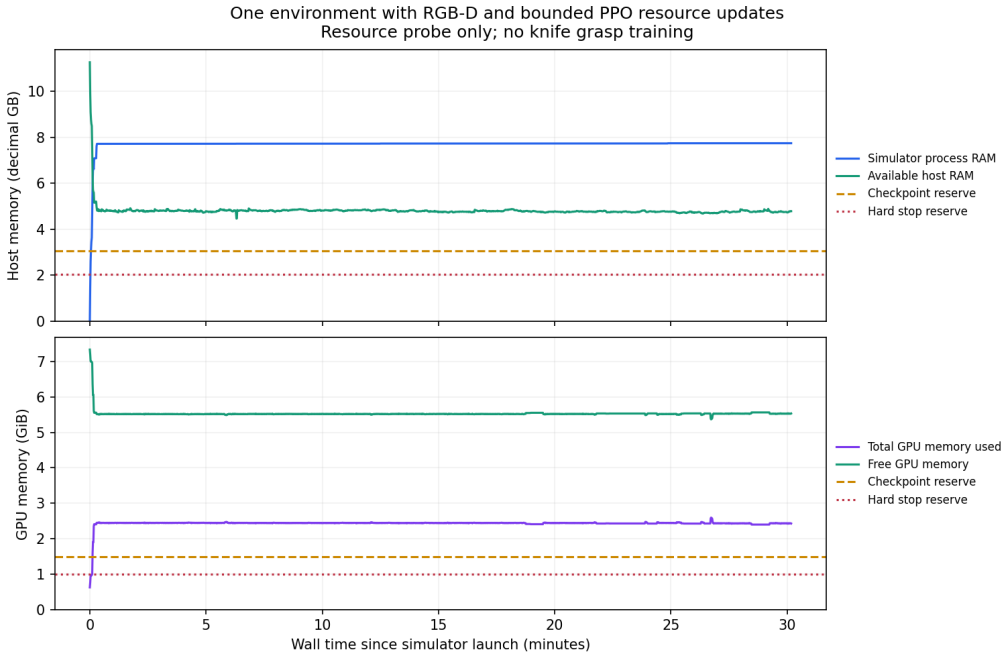
The proposed deployed actor may use images, joint feedback and recent action history. Exact object pose, damage variables and simulator contact identities may help rewards, debugging or a privileged training critic, but should not be silently supplied to a camera-based deployed actor.

A state-based teacher and a compact PPO policy were proposed as starting options. PPO is a method for updating a policy from trial outcomes; selecting it in a plan does not mean useful knife training has happened.

Current status: no trained knife-recognition system and no trained knife-pickup policy are established. A simulated RGB-D camera was exercised in a resource benchmark; that is evidence of camera operation and memory cost, not perception learning.

Evidence: [S02], [S04], [S05], [S06]. See the source index.

10 Auditing and benchmarking the RL stack



Saved Phase 2 resource plot. The image labels the workload as a resource probe, not knife-grasp training.

Phase 1 inventoried robot joints, software, CUDA capability and the proposed information/action contract. Physical sensor configuration and calibrated hardware limits remained open.

Phase 2 ran a reduced one-environment benchmark with robot, board, counter, free knife, RGB-D camera and a small untrained image encoder. The long run lasted 1,800.1 seconds, completed 277 resets and 12 tiny PPO updates, and exited normally.

Peak process RAM was 7.747 GB; minimum available RAM was 4.466 GB; peak total GPU use was 2,658 MiB. Measured action throughput was about 9.83 actions per real second. Checkpoint restoration and deliberately requested stop/resume paths were also checked.

Those 12 updates optimized a small posture workload. They did not train knife pickup or image recognition. A 30-minute stable run is not proof of multi-day stability or affordable full training.

Evidence: [S04], [S05], [S03]. See the source index.

11 Native integration and Phase 3

The original environment was preserved. A local .venv-native overlay inherited it and shadowed selected dependencies. Click, psutil, Numba/llvmlite, torchaudio and some utility packages were repaired or pinned for the tested stack.

The metadata check was not made universally clean: six known conflicts associated with explicit upstream version overrides remained documented. Targeted CUDA, compilation, audio, environment and adapter checks passed. This validates the exercised path, not every optional backend or importer.

What the native test actually connected

A native Isaac Lab DirectRLEnv used PhysX, native robot/object assets and an RSL environment wrapper. Physics ran at 240 Hz and control at 20 simulated Hz. A 13-action joint probe and 39-value observation tested the interface. This path had no optimizer or camera-based pickup policy.

A separate Phase 3 scene enabled self-collision, restricted exclusions to justified adjacent/fixed assemblies and corrected the ready posture. It used explicit provisional knife dynamics: 0.25 kg mass, 0.8/0.6 static/dynamic friction and zero restitution. These are assumptions, not measured knife properties.

Completed scoped checks	What remains open
28 knife-settling fixtures	A physical wrapped flat-centre pickup.
1,000 geometric resets	A detector validated against positive physical examples and images.
20 pre-approach IK checks	Complete collision-free routes and valid placement coverage.
Empty-hand/contact interface checks	Calibrated actuators, friction, sensors and hardware deployment.

Phase 3 is incomplete. The older edge-supported demo is not a positive example for this new scene, and full training has not been started.

Evidence: [S06], [S03]. See the source index.

12 Why flat central pickup is still difficult

The newer task starts the existing knife flat at the board centre, with its whole footprint inside the board. Fingers can no longer rely on a handle already hanging beyond the edge. Tilting or nudging through real simulated contact was allowed as an intermediate action.

Continuation	What was observed	What did not pass
Central approach	An opposed initial pinch could be acquired.	Retention through clearance and a firm wrap.
Seven tilt diagnostics	About 15 mm peak handle rise; an early supported plateau.	A firm unsupported lift.
Ten palm-seating diagnostics	About 36.8 mm handle rise with blade supported on the board.	Palm/three-finger wrap and unsupported hold.
Contact-preserving seating	Thumb/index pinch retained through a small adjustment.	Palm contact; lift was refused.
Finger-contact transfer	Brief ring contact reached 1.143 N for up to 0.4 s above 0.3 N.	Stable transfer to the other fingers and a wrapped hold.

Controller work included load-preserving goals, per-digit feedback, synchronized wrist/finger targets, tracking checks, ring clearance and bounded substep logging. Some modes exposed thumb-mount/board interference and stopped earlier rather than disabling inconvenient collisions.

Six transfer runs completed, while three additional launches were refused below the 10 GB startup reserve. These refusals were not physical tests. No attachment, duplicate knife or in-episode knife teleport was used to manufacture a pickup.

Raising the handle while the board still supports the blade is mechanically useful, but the board is helping carry the knife. It cannot satisfy an unsupported lift-and-hold claim.

Evidence: [S08], [S09], [S10], [S11], [S12]. See the source index.

13 Astra review and offline grasp refinement

While simulator work was paused, the Astra review separated three questions: can the mechanism form a useful final grip, why does the recorded pinch lose support, and is the contact/controller evidence reliable enough to explain the failure? It did not prove that the hand is incapable of grasping the knife.

The review noted abrupt index-contact loss within recorded physics substeps even when adjacent commanded targets were identical. Small tracking error did not guarantee retention. Missing subthreshold contact points and incomplete force/moment information limited causal conclusions.

Three subsequent offline checks

- Geometry and colliders: screened inner finger/palm regions and self-envelopes. Authored meshes are not necessarily the exact cooked collision hulls used by the physics engine.
- Controller replay: reproduced recorded targets and found that a prior projection residual did not describe the final command actually issued.
- Load support: checked whether candidate contacts can balance weight and its turning moment under provisional friction/actuator budgets.

Near-flat proposals were not accepted. Inner-wrap refinements brought selected surfaces close to the handle, but gaps, small intersections or failed load balance remained. Diagnostic logging was expanded to record raw contacts, separate friction points, actual link states and final-command kinematics.

A later 90-degree axial-roll proposal admitted conditional static support for 1.25 times the assumed knife weight and preserved the outward blade axis. It remained inactive: positive gaps, cooked geometry, retention and the route from a flat start were unresolved. It also lay outside the unchanged 20-degree in-place diagnostic limit.

An offline candidate is a proposal, not a new physical grasp success. The final diagnostic series reached 62 passing automated checks, while Phase 3 and real retention remained open.

Evidence: [S13], [S14], [S15], [S16], [S03]. See the source index.

14 Planning tomato slicing and considering FEM

You chose slicing as the next topic, with robot learning and control as the research objective. There was no tomato CAD model. The agreed first robot milestone was one slice of specified thickness, a known knife grasp and a clearly disclosed fixture holding the remainder. Later stages add the left hand, camera localization, repeated slices and learned pickup.

Option	What it offers	What it does not automatically solve
Reduced spring/particle model	A smaller mechanics prototype with inspectable forces and damage rules.	Realistic tomato anatomy, reliable contact or automatic fracture.
Coarse FEM	Volume elements and a continuum-material formulation.	Cutting: cracks still need a separation/contact method and calibration.
Detailed FEM	Potentially richer skin/flesh behaviour after measurement.	Cheap local training, correct parameters or hardware transfer.

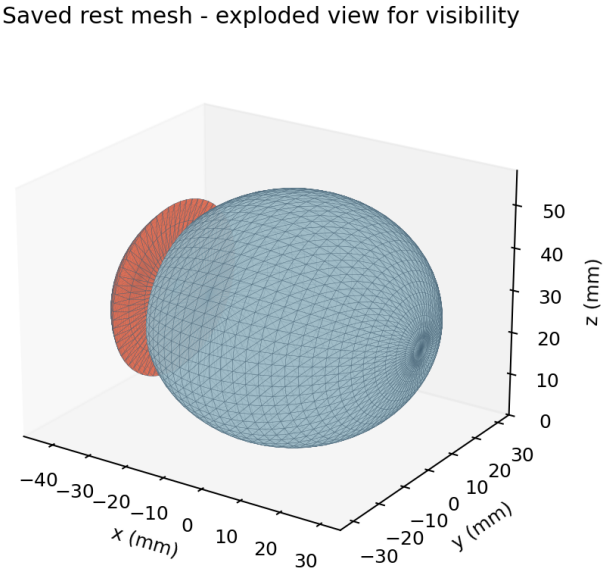
The offline capability audit inspected installed source and metadata. It recommended a reduced spring-based candidate using identified Newton building blocks, with coarse FEM as a comparison route. It found no demonstrated turnkey tomato slicing feature.

The robot was not migrated to Newton. The inspected simple solver lacked some mimic/equality and actuator-limit features needed by the existing hand. The later numerical tomato calculations also do not establish native Newton integration: they run as offline Python mathematics without loading the simulator.

A model that squashes is not automatically a model that cuts. A model that cuts along one predefined plane is not automatically able to make arbitrary or repeated slices.

Evidence: [S17], [S18], [S19]. See the source index.

15 The tomato has three representations



Red end slice is shifted left by 9 mm only in this illustration. No cutting or release is shown.

Actual saved refined mesh, illustrated with an artificial display offset. The visible gap is not a simulated separated slice.

- Visual geometry: what the tomato looks like. A smooth red shape alone has no mechanical meaning.
- Mechanical representation: points, springs, local volume protection and damage variables that determine forces.
- Collision/contact representation: closed surfaces used to stop the blade and cut faces passing through one another.

The synthetic tomato starts as an ellipsoid, 65 x 65 x 55 mm. The restricted benchmark proposes a 10 mm end slice. These dimensions describe a controllable test object, not a measured CAD reconstruction of a laboratory tomato. No detailed pulp, seeds or measured skin layer has been built.

Both pieces are represented separately at the planned cut, with distinct material identities. Intact interface mechanics hold them together initially. Later contact geometry uses 257 matching vertex pairs and 448 triangles on each cut face. A common position does not mean a common node ID.

The cutting interface is predefined. This makes the initial problem manageable, but it also tells the model in advance where material is allowed to fail. It is not a free-cutting material model.

Evidence: [S17], [S20], [S23], [S30]. See the source index.

16 Nodes, bonds and the damage-law correction

A node is a small material sample. A bond is a spring-like connection between samples. A patch groups several connections and owns an area of the future cut. Its damage variable D runs from 0 for intact to 1 for fully broken; an opening-history variable remembers the largest eligible opening reached.

The first 5 mm layout had 979 nodes, 10,510 springs, 403 registered cross-cut bonds and 47 interface patches. Removing all cross-cut bonds produced exactly two connected components. Leaving one intact connection kept the pieces connected. This was a bookkeeping test, not blade-driven separation.

The first important failure

A single-spring reference had the intended fracture energy, but copying its scalar damage coordinate into a five-spring bundle did not. Opening work was 13.60% too high and shear work 64.20% too low. Passing small software tests did not hide this failed mechanical gate.

The correction calculated the actual undamaged energy stored in all springs belonging to a patch. An equivalent opening coordinate was derived from that energy, so one patch retained one fracture-energy budget. The forces still came from the spring energy; no per-direction force multiplier was fitted to make the result pass.

All 47 actual patches then passed five loading directions each: opening, positive/negative y shear and positive/negative z shear. That is 235 cases. Complete failure removed cohesive stiffness and damping; unloading did not heal; ineligible contact did not accumulate damage.

The tested strength and fracture-energy values were synthetic engineering values. Energy consistency shows the equations are internally consistent in those cases, not that real tomato forces have been identified.

Evidence: [S20], [S21], [S22]. See the source index.

17 Closed surfaces, mass and spring stiffness

After damage, two pieces still need surfaces that can collide. A cloud of points can leave holes, and a visual shell can mistakenly stretch across a completed cut. Closed oriented triangle surfaces were therefore built separately for the slice and remainder.

The cut faces share matching positions and opposite outward normals, but no shared topological vertices. Surface connections and volume cells belong to one piece only; they must not secretly reconnect the slice after damage.

Making the added surfaces mechanical

The first closed-surface design expanded the network to 3,735 nodes. Added points received same-side connections. Their masses and stiffnesses then needed refinement: adding a pretty surface without accounting for its mass and force path would change the model unpredictably.

A fixed percentage of mass on the surface was replaced by volume-based nodal mass lumping. Temporary tetrahedral cells distributed volume and supported inertia checks. At that stage they were integration/diagnostic cells, not FEM force elements. The model was still spring-based.

Quantity in the documented refinement	Value / interpretation
Slice mass	7.737 g, recomputed from the faceted geometry.
Remainder mass	113.934 g; total synthetic mass was retained.
Small-strain checks	18 compression/shear cases passed.
Early explicit-step estimate	About 31 microseconds for the spring-only proposal; not a later validated dynamic timestep.

Positive directional spring weights were matched to a synthetic elastic response. Mass accounting and small-strain agreement were useful gates, but large deformation, contact and material calibration still had to be tested.

Evidence: [S23], [S24]. See the source index.

18 Finding collapse and adding volume protection

The first larger-deformation tests exposed a fundamental problem: local material cells could invert even though the force equations converged and the local energy minimum looked stable. An inverted cell has turned inside out; a converged answer can therefore still represent impossible geometry.

The spring-only model failed some compression paths as early as the 5% tested level. Only two of 18 planned paths completed their whole requested range. This was localized geometric failure, not a laptop crash or evidence that an entire tomato visibly collapsed.

What changed

The model became explicitly hybrid: springs plus local volume-protection energy. For each same-piece cell, J is current signed volume divided by rest volume. J=1 means unchanged local volume; J=0.5 means half; a nonpositive value means collapse or inversion.

The synthetic protection term is zero above J=0.5 and grows steeply as J approaches 0.1. Invalid trial states are rejected. The protection acts inside each piece only, so it does not create a hidden connection across the cut.

Protected-model test	Outcome
Compression	Reached 30% in the stated supported static cases.
Shear	Both signs through 50% in the tested directions.
Complete sweep	108 stages across 18 paths passed.
Smallest reported J	0.2224, above the 0.1 floor.

This was an actual model revision, not a looser acceptance rule. It is an uncalibrated anti-collapse regularizer, not proof that the model reproduces a real tomato's incompressibility or dynamic behaviour.

Evidence: [S25], [S26]. See the source index.

19 Why different mesh sizes gave different answers

A mesh is the chosen spacing and connectivity of material samples. A trustworthy model should not change drastically just because the discretization is moderately refined. The planned comparison target was 10% for the selected response measures.

Finding	Correction or outcome
Initial sensitivity	A 6 mm mesh changed slice compression reaction by 12.58%, above target.
Coincident 4 mm nodes	Same-side duplicate material/surface points were reused; opposite cut sides were not welded.
4 mm remainder inversions	Very thin local cells near free nodes were diagnosed and their geometry/attachments corrected.
Remaining slice disagreement	4-versus-5 mm thickness reaction still differed by 10.52%.
Rejected edge-only change	Adding cell edges alone worsened disagreement; it was not adopted as the answer.
Accepted slice refinement	Half-spacing interior samples plus cell-edge connections brought energy/reaction differences to 5.74% / 6.91%.

The same refinement rule was applied to every comparison mesh. The reference itself changed: the refined 5 mm response differed substantially from the old reference, by up to 22.97% in energy density and 31.13% in normalized reaction. Agreement was not achieved by pretending the old response stayed identical.

The later reference contains 4,032 nodes: a 1,195-node slice and 2,837-node remainder. This is the protected model used by the subsequent contact studies.

Meeting a 10% screen on selected paths is useful evidence of reduced sensitivity. It is not a universal continuum-convergence proof or material calibration.

Evidence: [S27], [S28], [S29], [S30], [S31]. See the source index.

20 Contact coupons and the full-node solver

Before combining everything, small contact coupons tested simple questions: can the surrogate blade load a patch, and can two finite cut outlines approach without passing through one another? These deliberately used restricted supports or rigid shapes.

The next solver included all 4,032 nodes, same-piece spring energy, volume protection, interface forces and unilateral normal contact. Unilateral means contact can push but cannot pull the surfaces together like glue.

Static equilibrium rather than a moving simulation

The offline solver searches for positions where applied contact loads and material forces balance. It does not advance a full dynamic scene with gravity, inertia, cameras and robot motors. An optimization guess is a proposed answer to the equations, not a physical frame of an accepted trajectory.

The first coupled stage passed 68 accepted checkpoints and 24 tests in its declared small-indentation and guided-reclosure scope. Damage was prescribed at 0, 0.5 or 1 for those diagnostics; it did not evolve because a blade cut material.

The initial sparse direct factorization exceeded the address-space budget and was rejected. A memory-bounded iterative method replaced it for that stage. The limits were not raised to label the failed calculation as successful.

The faces had fixed y/z guides and normal x motion. Those supports simplify the problem and must be disclosed. Passing this solver does not establish arbitrary 3D collision, a freely moving released slice, or contact with the real board and robot.

Evidence: [S31], [S32]. See the source index.

21 Fixing double-counted interface compression

The first coupled model still used finite-length bonds between interior nodes across the future cut. Under compression those bonds shortened and carried load, while face contact also carried load. The problem was not that all forces exactly doubled: the old intact interface carried 48.56% of the compressive load, and total actuator force was about 7.5% higher than the broken case.

Separating two physical jobs

- Cohesion resists opening and represents material still holding the two sides together.
- Contact resists compression and prevents the faces occupying the same space.
- Damage weakens cohesion, while contact must remain available even after complete breakage.

The correction replaced those interior cross-cut force paths with a zero-thickness surface-cohesion potential on matching face vertices. Normal compression contributes no cohesive normal energy. Positive opening and elastic shear can store cohesive energy, scaled by the patch's remaining integrity.

Each original patch retained one area/stiffness/fracture budget, distributed over a normalized eight-vertex support. This is a provisional transfer, not an unchanged finite-length bond law or a calibrated skin/flesh map.

Guided +0.05 mm reclosure	Corrected outcome
Intact, half-damaged and fully broken	Actuator/contact force agreed at about 0.017247 N.
Compressive cohesive contribution	Zero.
Validation	68 checkpoints, 60 tests, 235 patch/mode cases and 68 weak-contact stability checks passed.

The 403-bond registry remains important historical provenance, but the later contact model's active interface mechanics are surface cohesion. These are different model stages.

Evidence: [S33], [S32]. See the source index.

22 Letting blade contact cause partial damage

The next stage connected current blade contact to irreversible patch damage. A patch could grow damage only with a retained cutting-edge encounter, a current positive blade load and sufficient opening energy. Depth or elapsed time alone did not delete bonds.

The fixed-normal blade was a finite surrogate wedge. Its geometry and 8 mm bevel were not calibrated to a measured real cutting edge. Earlier records also contain different blade-envelope versus whole-knife dimensions; they should not be treated as interchangeable physical measurements.

At 4 mm insertion	Recorded result
Damaged patches	2 out of 47, both partial.
Maximum damage D	0.187862.
Complete interface failures	None.
Accepted saved states	27 full-node states across the scoped histories.
Other checks	74 tests and 94 supported blade/patch energy histories passed.

An energy ledger for fracture

Work delivered by the blade must be accounted for as recoverable material energy plus irreversible fracture loss. In the refined 3.5-to-4 mm interval, blade work was about 0.00112350 J, stored-energy increase 0.00082087 J and fracture dissipation 0.00030282 J. The residual was 0.062222% of fracture loss, below the 1% gate.

A positive tangent with damage frozen is not enough for a growing crack. This stage also included the loading-side softening correction in its local stability check. It still validated only a guided partial cut, not independent slice release.

Evidence: [S34]. See the source index.

23 Unloading, reclosure and no healing

After partial damage, we withdrew the blade rather than continuing immediately to a full cut. The purpose was to check two independent facts: the faces can close and contact correctly, and damaged material does not magically become intact when its gap closes.

Two branches started from the same accepted 4 mm partially damaged state. Withdrawal used 0.25 mm and 0.125 mm spacing. After natural closure, a disclosed back-cap support gently compressed, reopened and reclosed the slice. This support was a diagnostic aid, not a released-slice test.

Check across both branches	Result
Independent audited checkpoints	72.
Regression tests	82 passed.
Damage and maximum-opening history	Exactly unchanged from the partial-damage seed.
Largest face overlap	About 1.25e-14 m, below the 1e-9 m tolerance.
Compression contact load	About 0.017247 N under the small supported push.
Reopening	About 5.875 micrometres maximum gap under the small pull.

Unloading returned stored elastic energy, not previously spent fracture energy. The refined work error relative to initial elastic energy was about 0.162%. A separate small-cycle ledger prevented the tiny reclosure experiment from hiding errors behind a much larger historical fracture budget.

Closing a crack is not healing it. A damaged interface can touch and transmit compressive load while retaining reduced tensile stiffness. Complete separation and unrestricted post-release motion were not demonstrated.

Evidence: [S35]. See the source index.

24 The incomplete full-cut attempt

After the unloading checks, the evolving-damage model advanced from 4 mm to 5 mm insertion through eight accepted 0.125 mm checkpoints. Damage/history remained irreversible and the guided cut faces did not cross beyond tolerance.

At 5 mm in this earlier formulation	Result
Partially damaged patches	5.
Maximum D	0.448889.
Completely failed patches	0.
Material components	1: the intended slice was still attached.
Associated regression total	89 tests passed; a full cut did not.

The predefined cut face is about 39.688 mm high. An edge traverse across that height, and roughly 47.688 mm depth to present the full surrogate bevel at its bottom, is very different from a 5 mm shallow insertion.

Why the attempt stopped

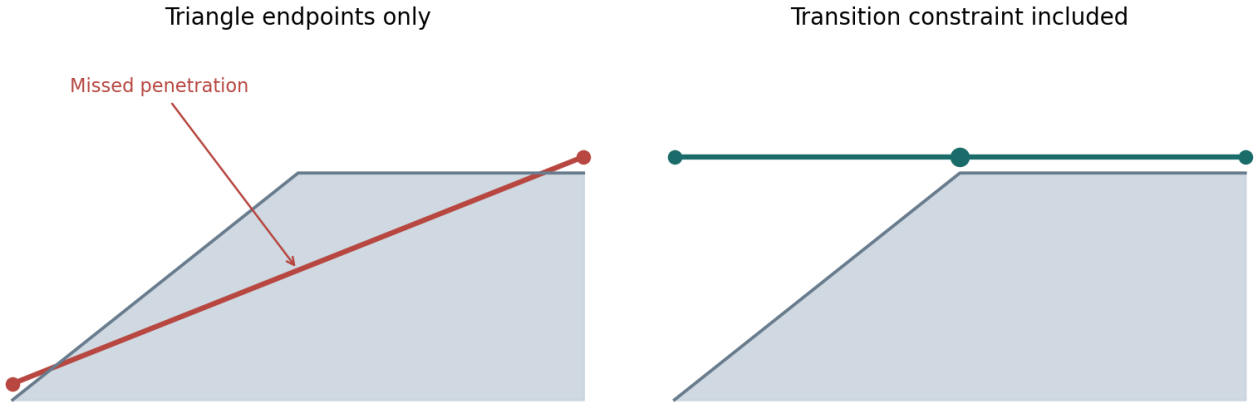
A geometry counterexample at 8.125 mm showed that original mesh vertices could all satisfy their blade constraints while triangles between them still intersected the capped bevel. Seven triangles exhibited the defect; maximum constructed penetration was about 57.966 micrometres. This was a synthetic feasible-at-vertices witness, not a crossing observed in an accepted dynamic cut.

Further shallow continuation was intentionally stopped after the in-flight 5 mm checkpoint. In that attempt the stop was a geometry qualification decision, not an exhausted resource cap. Later probes of the repaired solver hit computational limits; those are separate events.

Independent slice release remained unverified. Blade depth, a visible gap, or a passing software-test count could not turn this into a successful full cut.

Evidence: [S36]. See the source index.

25 Repairing contact between mesh vertices



Concept sketch: the red segment clears its endpoints but crosses the blade between them.

Concept illustration only. The marked interior point is a contact constraint, not a new free material node. The solver moves the original vertices.

Checking only the corners of a triangle can miss its interior crossing the blade. The new contact backend divides the geometry into the regions ahead of the finite edge, over the bevel and over the constant-thickness blade. Clearance is linear within each clipped region, so its boundary vertices cover the region.

Additional contact constraints are built at triangle-edge intersections with the physical tip and the bevel transition. Their positions are weighted combinations of the original mesh vertices. Those weights also distribute contact force back to the material.

The solver enforces the new constraints during optimization. A separate auditor clips the actual Cartesian triangles and checks clearance independently. Inactive placeholder rows preserve indexing but are excluded from the physical solve and carry zero load.

A separate swept-path check includes activation events and interior minima between saved times. One regression has safe endpoints but a 2.5 micrometre crossing mid-path; it is rejected. The work equation also includes the changing contact-coordinate weights, not only moving nodal bounds.

The correction is implemented in new offline workers. Historical nodal-contact runners are preserved for comparison; they are not approved entry points for a new full-cut claim.

Evidence: [S37]. See the source index.

26 What the latest checks establish

Latest check	Result and scope
Full cut-face reference at 8.125 mm	The saved 57.966 micrometre defect is resolved to about 3.47e-18 m numerical overlap. Uses disclosed quadratic supports, not nonlinear bulk tomato mechanics.
Protected tomato at 5 mm	A 4,032-node fixed-damage static equilibrium passes independent clearance, force balance, volume and stability checks.
Blade-work references	180/360 base-step studies pass the 1% gate; peak-relative errors fall from 0.833% to 0.417%.
Swept contact	Accepted reference paths stay within the 1e-9 m tolerance; crossing trial segments are discarded and subdivided.
Regression suite	103 tests pass: 14 new and 89 existing checks.
Full cut	Not retried or validated after this repair.

In the accepted 5 mm protected model, force residual is about 8.3e-12, minimum local volume ratio 0.454846 and normalized frozen-damage stability eigenvalue 0.001805. Peak worker RAM was about 185.68 MiB. Six entry-edge constraints were loaded; the shallow state did not load the bevel-cap constraints.

The deeper 5.25 mm nonlinear continuation exhausted the unchanged 45 s CPU cap. A larger 8.125 mm guess also produced a nonconvex free-space quadratic subproblem and was rejected. Neither result is an accepted mechanical state. The passing 8.125 mm supported surface reference must not be described as a passing bulk-tomato cut.

The contact geometry defect is corrected in scope. The remaining gates are deeper nonlinear convergence, coupling the new contact/work formulation to evolving damage, and genuine independent release.

Evidence: [S37], [S39], [S40]. See the source index.

27 What a future successful slice must mean

The first robot cutting benchmark remains a known grasp, a declared remainder fixture and one proposed 10 mm end slice. The fixture must not hold the slice itself. It is a simplification to isolate cutting, not a claim that the left arm already stabilizes the tomato.

Draft slicing check	Proposed meaning
Separation	No active material/attachment path remains; independent motion persists after blade withdrawal.
Thickness	Proposed 10 mm target with +/-2 mm tolerance, measured at multiple locations on the released material.
Remainder motion	Proposed translation <=5 mm and rotation <=5 degrees; deformation and fixture reaction reported separately.
Knife slip	For a robot-held knife, proposed additional limits of 2 mm / 3 degrees; all original grasp rules also remain.
Loads	Numerical force/torque limits are not yet confirmed; robot-level load acceptance remains blocked.
Left-hand clearance	A proposed 20 mm simulation margin, with uncertainty, is not hardware safety certification.

These slicing numbers are a draft, not a frozen calibrated hardware contract. The mechanical studies later defined specific numerical gates such as 1% work balance, 1e-9 m penetration tolerance, relative force residual 1e-6 and local volume ratio at least 0.1. Those gates are model-specific numerical checks, not real-world safety distances.

The frozen grasp contract remains separate: 5 cm unsupported lift, three-second hold, no more than 10 mm slip, palm/opposed-thumb support, three qualifying fingers with at least 115 degrees combined curl, and two inner contacts at least 15 mm behind their tips (or qualifying proximal contact). It has not been weakened to make a cut or grasp pass.

Evidence: [S17], [S38], [S07], [S13], [S37]. See the source index.

28 Control and RL after the mechanics gate

A conventional cutting controller is planned before claiming an RL improvement. It would use a bounded downward-and-drawing motion, with joint, speed, acceleration and effort limits. Cutting loads include sideways force and turning moments; merely supporting knife weight is not enough.

The sensor question matters

Wrist torque is provisionally assumed, not confirmed hardware. Torque is a turning effect. A full six-axis force/torque sensor measures three force and three torque components. Torque alone cannot generally reveal a unique cutting force, and estimated motor torque is not automatically a calibrated wrist measurement.

Camera calibration, tool/gravity compensation, sensor frames, timing, bias, limits and saturation must be verified. Force or admittance control should not be claimed before those prerequisites are established.

Planned learning stages

- Start with a small residual policy adjusting the bounded baseline rather than giving the policy unrestricted motion authority.
- Vary size, firmness, friction and blade properties within justified ranges; keep separate held-out evaluations.
- Replace exact state with camera estimates for a vision claim; include occlusion, delay and confidence.
- Replace the fixture with tested left-hand stabilization; then add repeated slices and changing remaining geometry.
- Integrate the separately evaluated pickup skill and test that its grasp survives cutting loads.

The actor must not receive hidden damage/connectivity labels as an undeclared answer. Such state can support rewards or evaluation. No full slicing or pickup training follows automatically from a passing mechanics test.

Evidence: [S17], [S02], [S04]. See the source index.

29 Where the work is stored

The active working project is different from the synced ChatGPT project mirror. The sources/ material in the mirror is read-only reference material. The actual robotics work described here is under:

`/home/omen/teamwork_projects/tomato_cutting_right_arm`

Location	What to look for
README.md	Historical scene/control demos and their limitations.
RL_VISION_KNIFE_PICKUP_PLAN.md	The phased camera/RL roadmap.
docs/rl/	Human-readable reports, criteria and provisional parameter files.
rl/	Offline numerical models, diagnostics, evaluators and test workers.
runs/rl/	Saved native simulator/benchmark experiments.
runs/offline/	Geometry, mechanical tests, source hashes and bounded-worker evidence.
logs/ and work/	Earlier scripted-demo captures, configuration and validation evidence.

For the latest contact work, begin with docs/rl/TOMATO_TRIANGLE_CONTACT_RESULTS.md and runs/offline/tomato_triangle_contact_01/. The final evidence names are validated5/, audit5_final.json, surface_final/, energy180_final.json, energy360_final.json, tests.json and final_validation.json.

Preliminary candidates and failed numerical attempts are intentionally retained. Use the final report and final_validation.json to distinguish accepted scoped results from superseded solver experiments. A candidate position file is not automatically a certified state.

Evidence: [S01], [S02], [S37], [S39]. See the source index.

30 Rerun the safe offline checks

These commands reproduce the latest offline checks without Isaac Sim or training. Use a terminal and run one worker at a time. They create a fresh result folder so the archived evidence is not overwritten.

```
cd /home/omen/teamwork_projects/tomato_cutting_right_arm
export OPENBLAS_NUM_THREADS=1
export OMP_NUM_THREADS=1
export MKL_NUM_THREADS=1
export PYTHONDONTWRITEBYTECODE=1
CHECK_DIR=$(mktemp -d "$PWD/runs/offline/manual_check_XXXXXX")
echo "$CHECK_DIR"
```

Run the 103-test suite

```
cp runs/offline/tomato_triangle_contact_01/run_tests.py \
  "$CHECK_DIR/run_tests.py"
.venv-native/bin/python "$CHECK_DIR/run_tests.py"
```

Expected ending: "Ran 103 tests" and "OK". The copied runner writes its summary in your new folder and retains the worker limits and lock. The overlay depends on the existing base environment; do not treat it as a portable standalone installation.

Reproduce the saved penetration correction

```
.venv-native/bin/python rl/offline_tomato_triangle_geometry.py \
  --output "$CHECK_DIR/blade_contact"
```

Look for passed: true, initial independent gap about -0.000057966 m, and final gap near zero. This is the supported surface reference, not full nonlinear tomato cutting.

Do not use the old full-kitchen or grasp GUI launcher as part of this offline procedure. Simulator work remains paused; these numerical commands do not relax the memory safeguards.

Evidence: [S37], [S40]. See the source index.

31 Rerun work accounting and audit a state

Continue in the same terminal so CHECK_DIR and the single-thread settings remain available.

Blade-work and swept-contact reference

```
.venv-native/bin/python rl/offline_tomato_triangle_energy.py \
  --steps 360 \
  --output "$CHECK_DIR/blade_energy.json"
```

Expected: passed: true and relative_peak_energy_error about 0.004167, meaning 0.417%, below 1%. This reversible reference has no tomato fracture or robot hand.

Recheck the protected tomato at 5 mm

```
.venv-native/bin/python rl/offline_tomato_triangle_contact.py \
  --depth-mm 5 \
  --guess runs/offline/tomato_triangle_contact_01/validated5/positions.npy \
  --output "$CHECK_DIR/tomato5"
```

This begins from an accepted position as an optimization guess. It independently solves/checks the known fixed-damage contact state. It is not a fresh cutting trajectory or a learned policy.

Independent geometry and equilibrium audit

```
.venv-native/bin/python rl/audit_tomato_triangle_contact.py \
  --checkpoint "$CHECK_DIR/tomato5" \
  --output "$CHECK_DIR/tomato5_audit.json"
```

Look for passed: true and flags: []. If only candidate.json exists, the worker did not establish an accepted equilibrium; do not interpret the candidate as success or feed it to the accepted-state audit. A lock error means another protected worker is active. Leave guard thresholds unchanged.

Evidence: [S37], [S39]. See the source index.

32 Reading results without misleading yourself

What you see	How to interpret it
passed: true	Read the scope: unit, surface reference, static contact or full task?
candidate.json / guess.npy	Uncommitted optimization output, not a physical step or accepted equilibrium.
summary.json	A worker summary; still check its passed flag, scope and independent audit.
source_hashes / state_hashes	Fingerprints connecting a result to the code and data used. A mismatch requires investigation.
time-limit / -9 worker return	A bounded process stopped. It does not establish task success or laptop failure.
tiny negative clearance	Compare with the frozen tolerance; do not silently call every negative value exactly zero.
D close to 1	Not fully broken. A remaining connection can still hold the pieces together.
103 tests pass	The latest selected suite passes; no full-cut or hardware success follows from the count.

For grasping, inspect actual contact locations, palm/inner-finger qualification, board support, hand-relative slip and the complete hold interval. For cutting, inspect material connectivity, energy balance, contact clearance, deformation validity and independent post-withdrawal motion.

Comparisons require consistent conditions: an edge-overhanging knife is different from a flat central knife; a self-collision-disabled demo is different from the later scene; a supported coupon is different from a free slice; a newer interface formulation needs its own consistent work ledger.

A useful personal tracking habit: write down the exact result folder, source version/hash, test scope, required checks, failures and next unresolved gate. Keep the old failed evidence instead of replacing it with a more flattering summary.

Evidence: [S03], [S07], [S36], [S37]. See the source index.

33 What should happen next

Tomato mechanics: the next bounded gate

- Make the nonlinear protected-model contact solve converge beyond 5 mm under the current memory/time limits. Inspect constraint-aware tangents and solver globalization; do not retune material simply to force a pass.
- Connect the new finite-triangle loads, moving-coordinate work and swept-path rejection to evolving damage. Start a consistent new undamaged trajectory and ledger.
- Demonstrate complete material disconnection, blade withdrawal, independent motion and fully damaged face reclosure. A visible opening is not enough.

Robot grasping: after resources and prerequisites

- Resolve cooked collision geometry, contact logging validation, terminal load-bearing wrap and a reachable flat-start route.
- Validate the unchanged grasp detector against real simulated positives and negatives before task learning.
- Recheck available RAM/VRAM after any hardware upgrade; keep one-environment experiments bounded.

Research and hardware

- Measure tomato dimensions, mass, ripeness and compression/cutting response; measure the knife edge and verify sensors.
- Establish a conventional control baseline and freeze a justified slicing contract before assessing RL improvements.
- Verify real interfaces, calibration and lab operating procedures before physical robot experiments.

Current decision: continue scoped offline development; no complete cut claim, no trained pickup/slicing policy and no demonstrated real-robot result. Native integration and training remain separate gates, not automatic consequences of a passing geometry test.

Evidence: [S17], [S37], [S03]. See the source index.

34 Beginner glossary

Term	Plain-language meaning
Actor / policy	The part of a learned controller that chooses actions from its observations.
Privileged critic	A training helper that may see extra simulator information unavailable to the deployed actor.
IK / FK	Inverse kinematics finds joint angles for a target; forward kinematics predicts a pose from angles.
DOF	An independent coordinate of motion; coupled follower joints are not extra independent commands.
Contact / cohesion	Contact pushes overlapping bodies apart; cohesion holds unbroken material together.
Coupon	A deliberately small supported example used to test one mechanical rule.
Quasistatic	Solve force balance without modelling the full motion history of inertia and acceleration.
Damage D / history	A variable for lost material integrity and a memory of prior eligible opening. Closing does not reset them.
Fracture energy	Irreversible work needed to create separated material area in the declared model.
Mesh / patch	The discretized material/surface samples; a patch groups interface area and its force/energy budget.
J / inversion	Local current-to-rest volume ratio; an inverted cell has an invalid signed material mapping.
Force residual	The remaining mismatch in force balance. Small residual is necessary but not sufficient.
Tangent / stability	How force changes with a small displacement; local stiffness checks do not prove global safety.
Barycentric weights	Fractions expressing a point on a triangle/edge as a combination of its original vertices.
Cooked collider	The collision geometry actually built by the physics engine, which can differ from a displayed mesh.
RAM / VRAM / RSS	Host memory, graphics memory, and a process's resident-memory measure.
Checkpoint / hash	A stored software state; a fingerprint used to detect changes in code or data.

Evidence: [S04], [S13], [S19], [S37]. See the source index.

Source index

Source paths below are relative to /home/omen/teamwork_projects/tomato_cutting_right_arm. They are local project evidence, not external literature or a new verification of the physical world.

[S01] Workspace history and scripted baselines
README.md

[S02] Vision-based RL plan
RL_VISION_KNIFE_PICKUP_PLAN.md

[S03] RL phase status and grasp history
docs/rl/PHASE_STATUS.md

[S04] Phase 1 model and software audit
docs/rl/PHASE1_AUDIT.md

[S05] Phase 2 resource benchmark
docs/rl/PHASE2_RESULTS.md

[S06] Native integration and Phase 3
docs/rl/NATIVE_INTEGRATION_AND_PHASE3.md

[S07] Frozen grasp criteria
docs/rl/phase3_grasp_criteria.json

[S08] Flat central grasp attempts
docs/rl/CENTRAL_GRASP_RESULTS.md

[S09] Supported in-place tilt
docs/rl/IN_PLACE_TILT_RESULTS.md

[S10] Palm seating
docs/rl/PALM_SEATING_RESULTS.md

[S11] Contact-preserving seating
docs/rl/CONTACT_SEATING_RESULTS.md

[S12] Finger-contact transfer
docs/rl/CONTACT_TRANSFER_RESULTS.md

[S13] Astra grasp review
docs/rl/ASTRA_GRASP_REVIEW.md

[S14] Offline geometry and controller audit
docs/rl/OFFLINE_GRASP_CHECK_RESULTS.md

[S15] Inner-wrap refinement and logging
docs/rl/INNER_WRAP_REFINEMENT_RESULTS.md

[S16] Knife mass, contact loads and rolled candidate
docs/rl/LOAD_SUPPORT_RESULTS.md

[S17] Tomato slicing implementation plan
docs/rl/TOMATO_SLICING_IMPLEMENTATION_PLAN.md

[S18] Installed capability audit
docs/rl/TOMATO_CAPABILITY_AUDIT.md

[S19] Reduced tomato prototype plan
docs/rl/NEWTON_TOMATO_PROTOTYPE_PLAN.md

[S20] First node/bond layout and failed bundle law
docs/rl/TOMATO_LAYOUT_AND_BOND_RESULTS.md

Source index - continued

Source paths below are relative to /home/omen/teamwork_projects/tomato_cutting_right_arm. They are local project evidence, not external literature or a new verification of the physical world.

[S21] Energy-consistent patch correction
docs/rl/TOMATO_PATCH_ENERGY_RESULTS.md

[S22] All 47 patch energy checks
docs/rl/TOMATO_ALL_PATCH_ENERGY_RESULTS.md

[S23] Closed cut-face geometry
docs/rl/TOMATO_CLOSED_SURFACE_RESULTS.md

[S24] Surface mass and stiffness
docs/rl/TOMATO_SURFACE_MECHANICS_RESULTS.md

[S25] Large deformation failures
docs/rl/TOMATO_LARGE_DEFORMATION_RESULTS.md

[S26] Local volume protection
docs/rl/TOMATO_VOLUME_PROTECTION_RESULTS.md

[S27] Volume and mesh sensitivity
docs/rl/TOMATO_VOLUME_SENSITIVITY_RESULTS.md

[S28] Coincident-node repair
docs/rl/TOMATO_4MM_MESH_COMPARISON_RESULTS.md

[S29] Remainder inversion correction
docs/rl/TOMATO_4MM_REMAINDER_REFINEMENT_RESULTS.md

[S30] Slice mesh refinement
docs/rl/TOMATO_SLICE_MESH_REFINEMENT_RESULTS.md

[S31] Local blade and cut-face coupons
docs/rl/TOMATO_CONTACT_COUPON_RESULTS.md

[S32] First full-node contact solver
docs/rl/TOMATO_COUPLED_CONTACT_RESULTS.md

[S33] Compression/interface model correction
docs/rl/TOMATO_INTERFACE_COMPRESSION_RESULTS.md

[S34] Blade-coupled partial damage
docs/rl/TOMATO_BLADE_DAMAGE_RESULTS.md

[S35] Partial-damage unloading and reclosure
docs/rl/TOMATO_UNLOADING_RECLOSURE_RESULTS.md

[S36] Incomplete full-cut qualification attempt
docs/rl/TOMATO_FULL_CUT_RESULTS.md

[S37] Finite-triangle contact correction
docs/rl/TOMATO_TRIANGLE_CONTACT_RESULTS.md

[S38] Draft slicing acceptance contract
docs/rl/slicing_criteria.json

[S39] Latest machine-readable qualification status
runs/offline/tomato_triangle_contact_01/final_validation.json

[S40] Latest 103-test result
runs/offline/tomato_triangle_contact_01/tests.json